

PostalMS

Sprint Planning Document

CST2550 | The Leftovers | Middlesex University | March 2026

3 Sprints | 3 March 2026 - 9 April 2026 | 43 Tasks Total

All 43 Tasks Completed

Project Overview

PostalMS was developed using the SCRUM agile methodology. The project was divided into three sprints each lasting approximately one week. Each sprint had a defined set of tasks assigned to specific team members with clear priorities. Sprint planning was conducted during team meetings held in the Ritterman Building and Sheppard Library at Middlesex University.

Team Members

- Durga Prasad Avala (M01038074) — Team Leader and Developer (SCRUM Master)
- Meet Sanjay Mande (M01000862) — Secretary
- Tapan Keyur Bhai Patel (M01050963) — Developer
- Forhad Hossain (M01044942) — Tester

Priority Key

- High — Must be completed for the application to function or for submission requirements
- Medium — Important but can be completed after High priority tasks
- Low — Administrative tasks that support the project

Sprint 1

Project Setup and Data Structure Design

Sprint Number	Sprint 1 of 3
Start Date	Monday 3 March 2026

End Date	Sunday 9 March 2026
Sprint Goal	Set up the GitHub repository, create the initial project structure, design all three custom data structures and assign team roles
Total Tasks	10
Completed	10
Status	Completed

Sprint 1 Backlog

ID	Task	Assigned To	Priority	Status
S1-01	Create GitHub repository with correct folder structure	Durga	High	Completed
S1-02	Set up C# WinForms project and push initial solution to repo	Durga	High	Completed
S1-03	Design CustomHashTable class with chaining collision resolution	Durga	High	Completed
S1-04	Design CustomBST class with recursive insert and search	Durga	High	Completed
S1-05	Research and plan CustomQueue linked list implementation	Tapan	High	Completed
S1-06	Create class diagram in draw.io and push to design folder	Meet	Medium	Completed
S1-07	Create ERD diagram in draw.io and push to design folder	Meet	Medium	Completed
S1-08	Set up MSTest project referencing main application	Forhad	Medium	Completed
S1-09	Review CST2550 coursework brief and identify all requirements	Forhad	Medium	Completed
S1-10	Set up meetings folder and document Meeting 1 minutes	Meet	Low	Completed

Sprint 1 Review

All 10 tasks were completed successfully within the sprint. The GitHub repository was created with the correct folder structure. The initial C# WinForms project was pushed to the repository. The design for all three custom data structures was agreed and documented. The class diagram and ERD were created in draw.io. The MSTest project was set up and confirmed working.

Sprint 1 Retrospective

- **What went well:** Repository setup was completed quickly. All team members understood the project requirements clearly after reviewing the brief together.
- **What could be improved:** More detailed task breakdown would have helped estimate time better for the implementation sprint.
- **Action for next sprint:** Break down implementation tasks into smaller sub-tasks with clearer deadlines.

Sprint 2

Implementation of Data Structures, Database and Core UI

Sprint Number	Sprint 2 of 3
Start Date	Monday 10 March 2026
End Date	Sunday 16 March 2026
Sprint Goal	Implement all three custom data structures from scratch, complete the SQL database script with sample data, and build the core UI pages
Total Tasks	14
Completed	14
Status	Completed

Sprint 2 Backlog

ID	Task	Assigned To	Priority	Status
S2-01	Implement CustomHashTable with Add, Search, Delete, Resize, Clear	Durga	High	Completed
S2-02	Implement CustomBST with Insert, Search, InOrder, FindMin, Delete	Durga	High	Completed
S2-03	Implement CustomQueue with Enqueue, Dequeue, Peek, ToList, Clear	Tapan	High	Completed
S2-04	Write complete SQL database script with Users, Parcel, Delivery	Durga	High	Completed
S2-05	Create all stored procedures for login, registration and parcels	Durga	High	Completed
S2-06	Add sample data including 3 user accounts and 5 sample parcels	Durga	High	Completed
S2-07	Implement LoginForm and RegisterForm with database connection	Durga	High	Completed
S2-08	Implement MainForm with navigation bar and page switching	Durga	Medium	Completed
S2-09	Assist with WinForms UI layouts and red colour theme styling	Tapan	Medium	Completed

ID	Task	Assigned To	Priority	Status
S2-10	Write unit tests for Hash Table operations (12 tests)	Forhad	High	Completed
S2-11	Write unit tests for Queue operations (10 tests)	Forhad	High	Completed
S2-12	Write pseudocode for Hash Table with step explanations	Meet	Medium	Completed
S2-13	Begin drafting PDF report introduction and design sections	Meet	Medium	Completed
S2-14	Document Meeting 2 minutes and upload to meetings folder	Meet	Low	Completed

Sprint 2 Review

All 14 tasks were completed successfully. The CustomHashTable, CustomBST and CustomQueue were all implemented from scratch without using any standard template library collections. The SQL database script was completed with all three tables, stored procedures, views and sample data. The LoginForm and MainForm were implemented and connected to the database. Unit tests for the Hash Table and Queue were written and confirmed passing.

Sprint 2 Retrospective

- **What went well:** All three data structures were implemented on time. The database script was comprehensive and covered all required functionality.
- **What could be improved:** The UI implementation took longer than expected due to the complexity of the WinForms layout system.
- **Action for next sprint:** Prioritise completing all UI pages early so testing can begin sooner.

Sprint 3

UI Completion, Unit Testing, Report, Video and Submission

Sprint Number	Sprint 3 of 3
Start Date	Monday 17 March 2026
End Date	Thursday 9 April 2026
Sprint Goal	Complete all remaining UI pages, finish all 59 unit tests, complete the pseudocode and PDF report, record the video and submit everything
Total Tasks	19
Completed	19
Status	Completed

Sprint 3 Backlog

ID	Task	Assigned To	Priority	Status
S3-01	Implement ParcelsView with send form and parcel dashboard	Durga	High	Completed
S3-02	Implement DeliveriesView with tracking and status updates	Durga	High	Completed
S3-03	Implement HomeView, HelpView and InfoView pages	Durga	Medium	Completed
S3-04	Implement ProfileView with edit profile and photo upload	Durga	Medium	Completed
S3-05	Implement offline AI assistant with parcel tracking and price logic	Durga	High	Completed
S3-06	Write unit tests for BST operations (13 tests)	Forhad	High	Completed
S3-07	Write unit tests for price calculation (9 tests)	Forhad	High	Completed
S3-08	Write unit tests for tracking ID validation (6 tests)	Forhad	High	Completed
S3-09	Write integration tests for all three data structures (5 tests)	Forhad	High	Completed
S3-10	Run all 59 unit tests and confirm all pass in Visual Studio	Forhad	High	Completed
S3-11	Complete pseudocode for BST and Queue with time complexity tables	Meet	High	Completed
S3-12	Complete PDF report design, testing and conclusion sections	Meet	High	Completed

ID	Task	Assigned To	Priority	Status
S3-13	Assist with UI form layouts and final styling adjustments	Tapan	Medium	Completed
S3-14	Record MP4 video demonstration under 6 minutes	Tapan	High	Completed
S3-15	Final check of all repo folders and confirm everything is uploaded	Forhad	High	Completed
S3-16	Push all source code and test project to GitHub	Durga	High	Completed
S3-17	Invite all 4 markers to GitHub repository as collaborators	Durga	High	Completed
S3-18	Document Meeting 3 and Meeting 4 minutes and upload to repo	Meet	Low	Completed
S3-19	Submit .txt file with GitHub URL before deadline	Durga	High	Completed

Sprint 3 Review

All 19 tasks were completed successfully. All remaining WinForms pages were implemented. The offline AI assistant was implemented as a bonus feature. All 59 unit tests were written and confirmed passing in 66 ms with 0 failures. The pseudocode document was completed. The PDF report was finished. The MP4 video was recorded. All files were pushed to GitHub and all 4 markers were invited as collaborators.

Sprint 3 Retrospective

- What went well: All 59 unit tests passed on the first run with no failures. The AI assistant bonus feature was completed successfully.
- What could be improved: Report writing could have started earlier in Sprint 2 to reduce time pressure in Sprint 3.
- Action for future projects: Allocate dedicated time for documentation from the beginning rather than leaving it to the final sprint.

Project Summary

Total Sprints	3
Total Tasks	43
Tasks Completed	43
Tasks Incomplete	0
Unit Tests Written	59
Unit Tests Passed	59
Unit Tests Failed	0
Project Duration	3 March 2026 - 10 April 2026
Submission Deadline	Friday 10 April 2026

All 43 tasks across all 3 sprints were completed successfully before the submission deadline. The PostalMS application was built using C# WinForms and Microsoft SQL Server with three custom data structures implemented from scratch. All 59 unit tests passed. The PDF report, pseudocode document, meeting minutes, class diagram, ERD and video demonstration were all completed and uploaded to the GitHub repository.